

ITERATION 3 / C0

Iteration 2 Legacy Items

Closure, public smoke test, screenshots, and QA runtime evidence

AC9.4.1 PASS	9 legacy checks recorded	28 browser captures	2 embedded QA files
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Release identity

Repository branch	main
Application SHA tested in production	368212375a25b3443ac002e7526a657dca2e35f2
Render frontend	team04-marine-observation-frontend - deploy dep - dar23bjbc2fs7381mq30 - live 2026-09-25 07:24:58 UTC
Render API	team04-marine-observation-api - deploy dep - dar23bjbc2fs7381mq1g - live 2026-09-25 07:27:35 UTC
Iteration 2 baseline	iteration2-final - 38b88ec50c362e40b7f1444fc54f9d35ba61e37a

Purpose and scope

This single-file evidence pack records the Iteration 2 C0 gaps carried into Iteration 3, the fixes, local synthetic-data retests, and the public Render release-candidate smoke. Public smoke used read-only application flows apart from the beach page automatic species-distribution prediction request. It did not submit reports, cleanups, or restore requests.

The live application SHA is the code commit deployed to Render. Later repository commits in this evidence update contain documentation and QA assets only; they did not change the application tested in the smoke run.

1. Iteration 2 legacy items and fixes

AC2.4.4 - Cleanup history

Gap Cleanup confirmation implied that the source report had been deleted, although the backend retained it.	Fix Updated the confirmation to say the source report is kept as history and is no longer counted. Backend deletion behavior remains unchanged.
Retest evidence Local synthetic SQLite run: cleanup returned HTTP 201; GET /reports/mine returned HTTP 200 and retained the resolved report; the beach response counted two remaining reports. The public cleanup mutation was not repeated.	Result PASS

AC3.3.3 - Active freshness

Gap Resolved or older-than-90-day reports could determine the displayed latest-report date.	Fix Added newestCountedReportAt, excluding resolved and expired reports. The beach page displays the active timestamp. lastReportedAt remains historical; latestContributingReportAt remains a compatibility alias.
Retest evidence Local: HTTP 200, two eligible reports, newestCountedReportAt=2026-09-23T22:30:31.805426+08:00; the resolved target and 2026-06-07 report did not drive freshness. Public: GET /beaches/morib returned HTTP 200 with four eligible reports and newestCountedReportAt=2026-09-18T14:12:29.080315+08:00; UI displayed NEWEST COUNTED - 2026-09-18.	Result PASS

Terminology

Gap /method omitted the 90-day median; /method/ai contradicted Small=1; map and beach wording differed; Very high and Severe were inconsistent.	Fix Documented the median/window and minimum-report rule; clarified Small-only evidence versus post-cleanup Small; standardized INSUFFICIENT DATA; explained participant-facing Very high versus internal API Severe.
Retest evidence Local and public pages were visually checked. Public /method and /method/ai rendered; GET /scoring-method returned HTTP 200.	Result PASS

D1 - Invalid beach and offline

Gap Invalid beach and network failures collapsed into a generic load error without a useful recovery action.	Fix Invalid IDs show Beach not found and Back to map. Offline failures show You seem to be offline and Retry.
Retest evidence Public GET /beaches/not-a-real-beach returned HTTP 404 and the not-found UI appeared. Under offline emulation, beach reads failed with ERR_INTERNET_DISCONNECTED; after reconnection and Retry, GET /beaches/morib returned HTTP 200 and the page recovered.	Result PASS

F1 - Cold start

Gap A sleeping free-tier service could leave a blank page.	Fix Added the HTML boot shell Waking up the server... and read-only scripts/prewarm_demo.py requests to /health, /beaches, and /cleanup-events. No write retry was added.
Retest evidence Local slow-network emulation used 400 ms latency and 50 KB/s download. The shell appeared while the app chunk was blocked; after unblocking, the app became ready, no POST was observed during startup, and the ready-state console had no warnings/errors. A sleeping Render cold start was not separately simulated.	Result PASS

Beach severity card

Gap Hero imagery or gradient could show through the overlapping status card.	Fix Made the overlapping status card opaque.
Retest evidence Local: Morib and Remis at 390 x 844 and 320 x 844. Public responsive smoke: Morib at both sizes.	Result PASS

D3 - Unfinished draft

Gap Browser confirm could discard a draft without an in-app choice.	Fix Added an in-app dialog with Continue draft and Discard draft & start new.
Retest evidence Local screenshot shows both explicit choices; no JavaScript confirm dialog was present.	Result PASS

D5 - Duplicate report

Gap The warning exposed a raw report ID and lacked useful context.	Fix Warning now identifies the report by date and beach.
Retest evidence Local same-day Pantai Morib warning showed the date and beach and did not expose a raw report ID.	Result PASS

D7 - Participant ID

Gap Restore participant ID accepted non-digits or incorrect lengths.	Fix Input is limited to exactly four digits before restore.
Retest evidence Local input 12a34567 was reduced to 1234. No restore was submitted.	Result PASS

2. Public release-candidate smoke

Smoke was performed in the Codex in-app browser against the Render frontend and API on 2026-09-25. A direct API health request returned HTTP 200 with the body below.

```
{"database":"configured","status":"ok"}
```

Flow	Status	Observed evidence
Frontend /map; GET /beaches; GET /auth/me	200	Map rendered; beach list and session reads completed.
GET /beaches/morib	200	Four valid/eligible reports; active newestCountedReportAt was 2026-09-18T14:12:29.080315+08:00.
GET /cleanup-targets/morib; GET /beaches/morib/cleanups/latest; GET /scoring-method	200	Supporting read APIs responded.
Frontend /method; /method/ai	200	Both pages rendered; scoring-method API returned 200.
GET /beaches/not-a-real-beach	404	UI showed Beach not found and Back to map. Related invalid-beach cleanup reads also returned 404.
Offline beach read	Network error, then 200	Offline emulation produced ERR_INTERNET_DISCONNECTED; Retry after reconnect restored Morib details.
POST /api/species-distribution/predict	200	Automatic page inference request. It did not create a report or cleanup.

Public writes: no POST /reports, POST /cleanups, or restore submission was observed. Mutation-oriented checks used local synthetic data.

Browser console: 0 warnings and 0 errors in the public smoke query. **Render API error logs:** 0 entries from 2026-09-25 07:24:00 UTC through 07:37:35 UTC.

Warnings recorded: API startup logged an ONNX Runtime GPU-device detection warning because /sys/class/drm/card0 was unavailable. Health and prediction endpoints returned HTTP 200; GPU inference was not asserted. A separate local Vite development-server session logged two non-blocking messages: React did not recognize the fetchPriority DOM prop, and a [scoring] diagnostic compared equal values with different object-key order. They were absent from the local production-preview ready load and public RC console query.

Network and console evidence is a sanitized CDP summary; a HAR export was not saved.

3. Local run and CI evidence

Local UI and API testing used an isolated SQLite database populated with synthetic data. The cleanup completion request returned HTTP 201, the retained report remained visible through GET /reports/mine (HTTP 200), and the resolved item was no longer counted. Active freshness was tested with a resolved report and an older-than-90-day report present. Invalid-beach API returned HTTP 404; offline emulation recorded ERR_INTERNET_DISCONNECTED and displayed Retry.

D3, D5, and D7 were exercised locally. The D7 recovery token was left blank and no restore was submitted. F1 used a local production preview with throttled network and a temporarily blocked app chunk; it does not claim a public Render cold-start simulation.

GitHub Actions run 35860555400 passed frontend typecheck, tests, production build, and the backend suite before deployment. The stale frontend expectation following the NO DATA to INSUFFICIENT DATA terminology change was corrected.

Retained Iteration 2 decisions: write retries (F1), flow bottom navigation (F2), keep/edit logging (F4), duplicate-rule changes (F8), and notebook font sizing (F11) remain unchanged. AC9.1.4 was not reopened for SQLite; PostgreSQL concurrency remains covered by AC10.1.2.

4. QA runtime files embedded in this PDF

The PDF embeds the sanitized SQLite fixture and the local screenshot receiver as downloadable attachments. Open the PDF attachment pane to extract them; this keeps the executable/test data bundled with the human-readable evidence.

Embedded file	Size and SHA-256	Contents / use
qa-fixture.sqlite	147,456 bytes efdf17d596c085ea8f3a79187470af07c725fb58b0984b3a0478b59fb09c1f83	Four synthetic reports and one cleanup action. Participant ID is 0000; local token is a non-production placeholder; report-photo records were removed. SQLite integrity_check returned ok.

Embedded file	Size and SHA-256	Contents / use
receive_screenshot.py	1,201 bytes f6151dd43be473e3526d5df19e055f941d7bdea3ac6d8efea952b16101277522	Python standard-library helper on 127.0.0.1:5174; GET /health and bounded JPEG POST /capture/.jpg; writes captures to the evidence screenshot folder.

The original local working database is intentionally not embedded: it contains its generated auth token and report-photo data. The attached fixture is sanitized and integrity-checked.

Screenshot receiver source

```
from http.server import BaseHTTPRequestHandler, HTTPServer
from pathlib import Path
import re

OUTPUT_DIR = Path(__file__).resolve().parents[1] / "screenshots"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

class Handler(BaseHTTPRequestHandler):
    def do_GET(self):
        if self.path != "/health":
            self.send_error(404)
            return
        self.send_response(200)
        self.end_headers()
        self.wfile.write(b"ok")

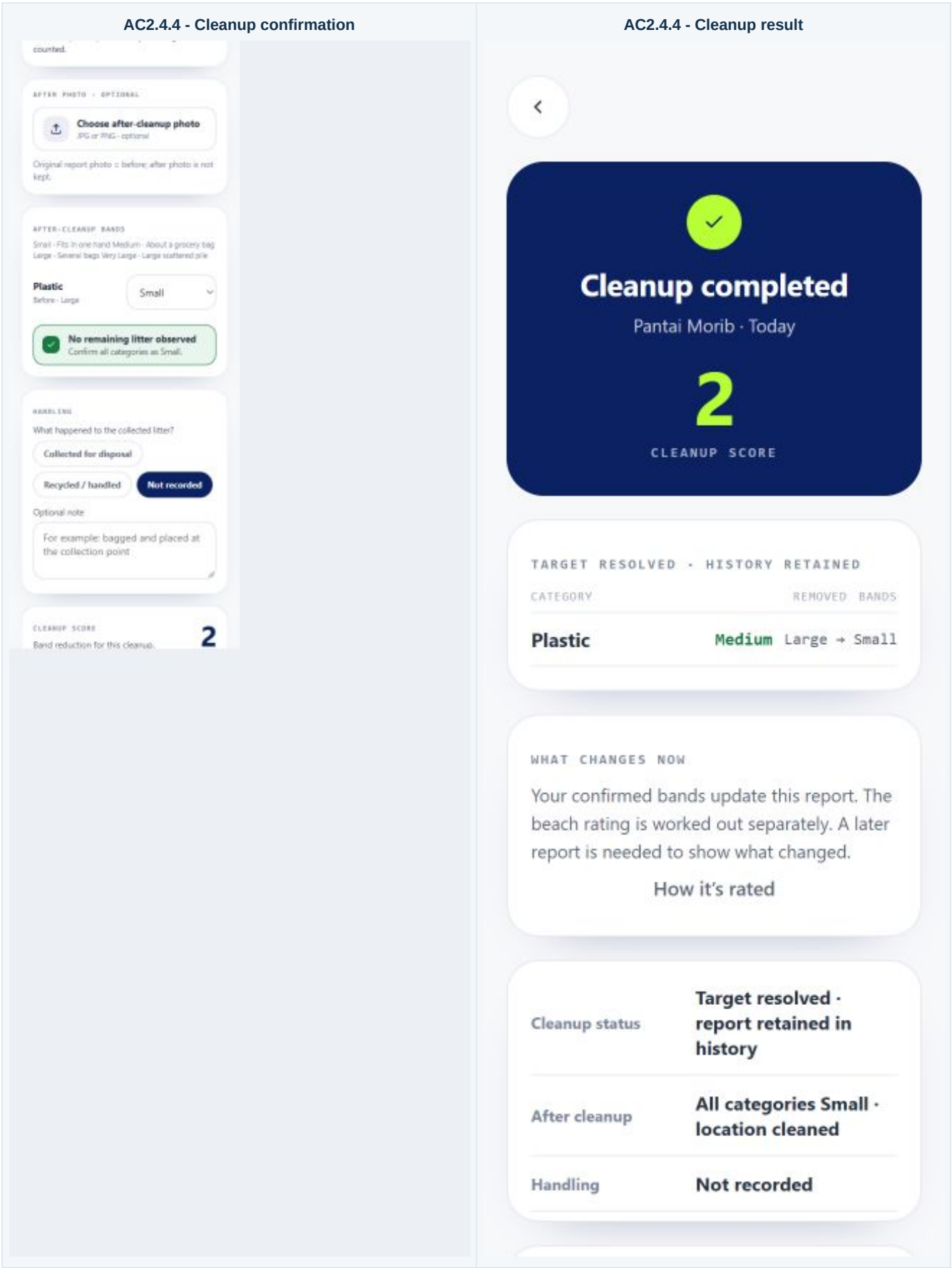
    def do_POST(self):
        name = self.path.removeprefix("/capture/")
        if not re.fullmatch(r"[a-z0-9-]+\.\jpg", name):
            self.send_error(400)
            return
        length = int(self.headers.get("Content-Length", "0"))
        if length <= 0 or length > 15_000_000:
            self.send_error(413)
            return
        payload = self.rfile.read(length)
        if not payload.startswith(b"\xff\xd8\xff"):
            self.send_error(415)
            return
        (OUTPUT_DIR / name).write_bytes(payload)
        self.send_response(201)
        self.end_headers()
        self.wfile.write(name.encode())

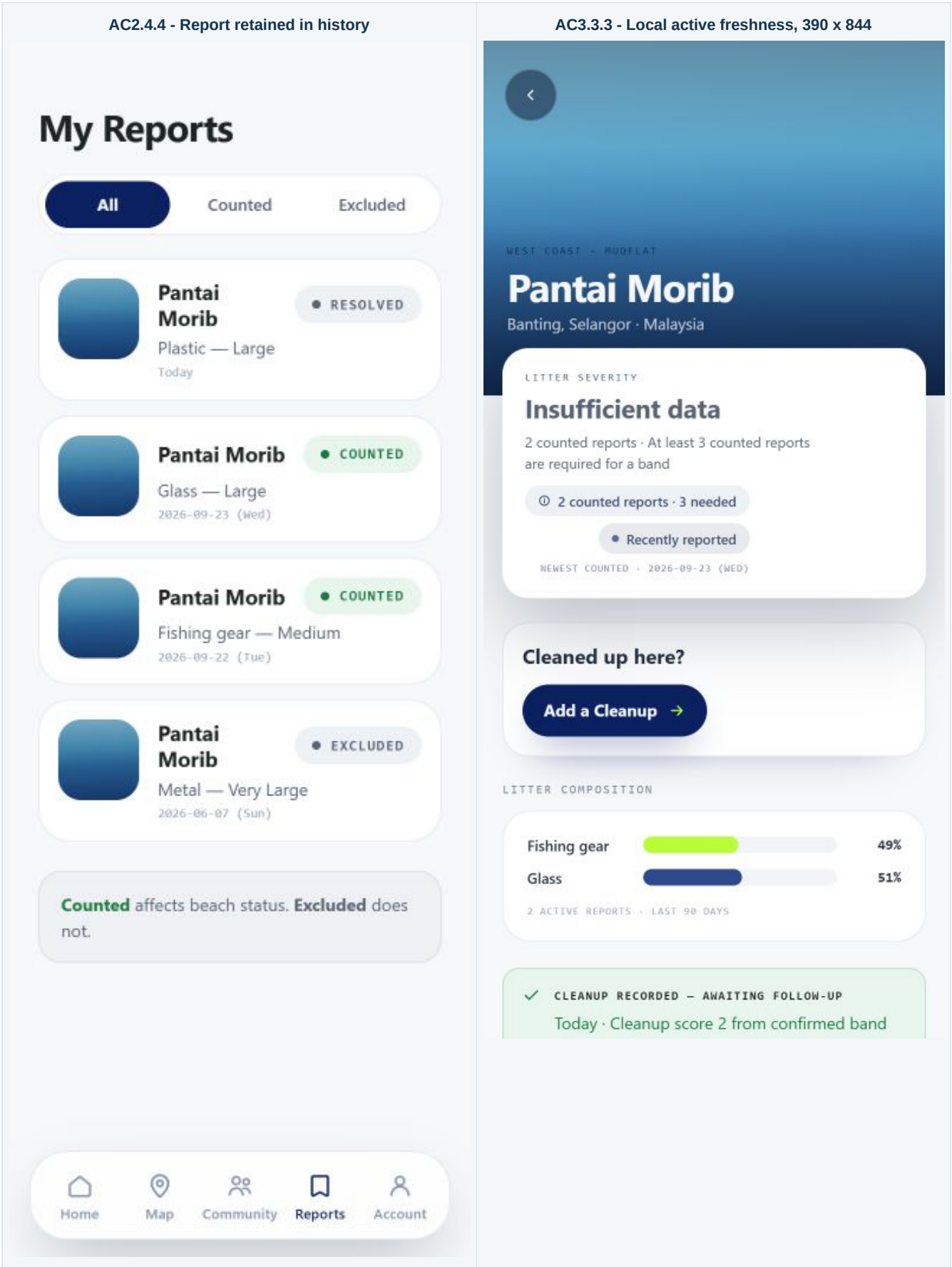
    def log_message(self, format, *args):
        return

HTTPServer(("127.0.0.1", 5174), Handler).serve_forever()
```

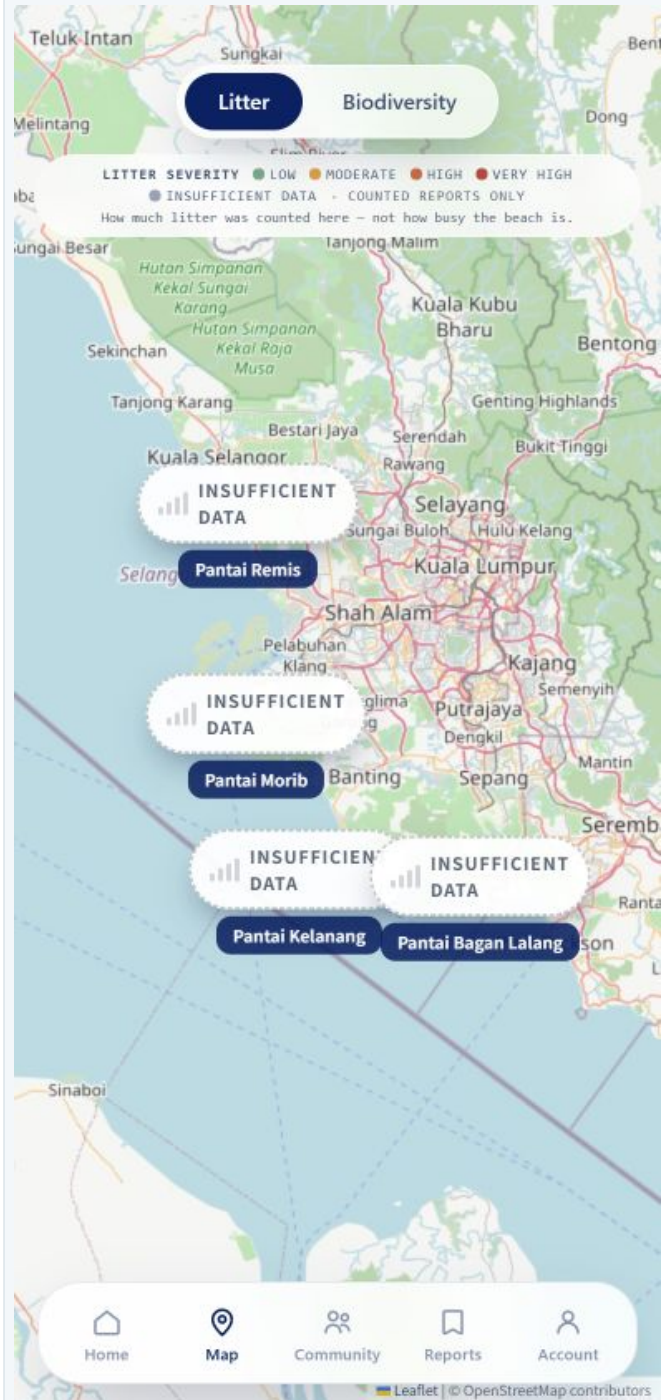
5. Browser screenshot gallery

Local synthetic-data and deployed public release-candidate captures. Viewport sizes are shown in each caption.

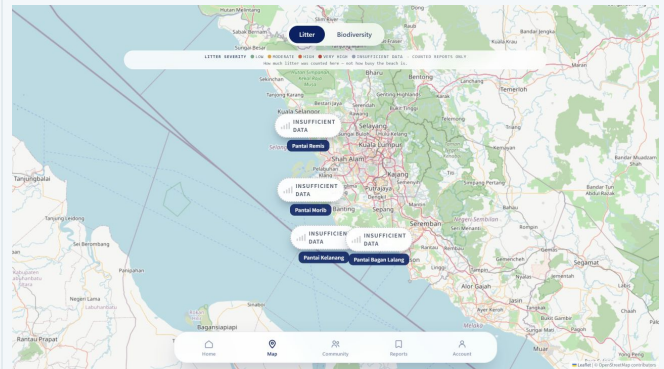




Terminology - Map, 390 x 844



Terminology - Map capture



Terminology - Method page

SCORE FOR ALL BEACHES

How the severity band is decided

Fixed litter rules. AI only suggests.

THE RULE

HOW CATEGORY

category weight × quantity level

HOW REPORT

highest category score

HOW BAND

median of active counted report scores - latest 90 days

CATEGORY		QUANTITY	
Fishing gear	1.00	Small	1
Plastic	0.85	Medium	2
Glass	0.70	Large	3
Metal	0.60	Very Large	4
Other	0.50		
Paper	0.35		

BANDS

Low

below 1.5

Moderate

1.5 - <2.5

Terminology - AI method page

MODEL PERFORMANCE

How the AI suggestion works

A suggestion is never submitted on its own. You confirm it first.

HOW IT WORKS

- You take a photo.
- AI suggests a category and size.
- Confirm or correct both.

TRAINED WITH OPEN WASTE-LITTER PHOTOS FROM SEA AND TACO

WHAT IT CAN SUGGEST

Plastic

Fishing gear

Glass

Metal

Paper

Other

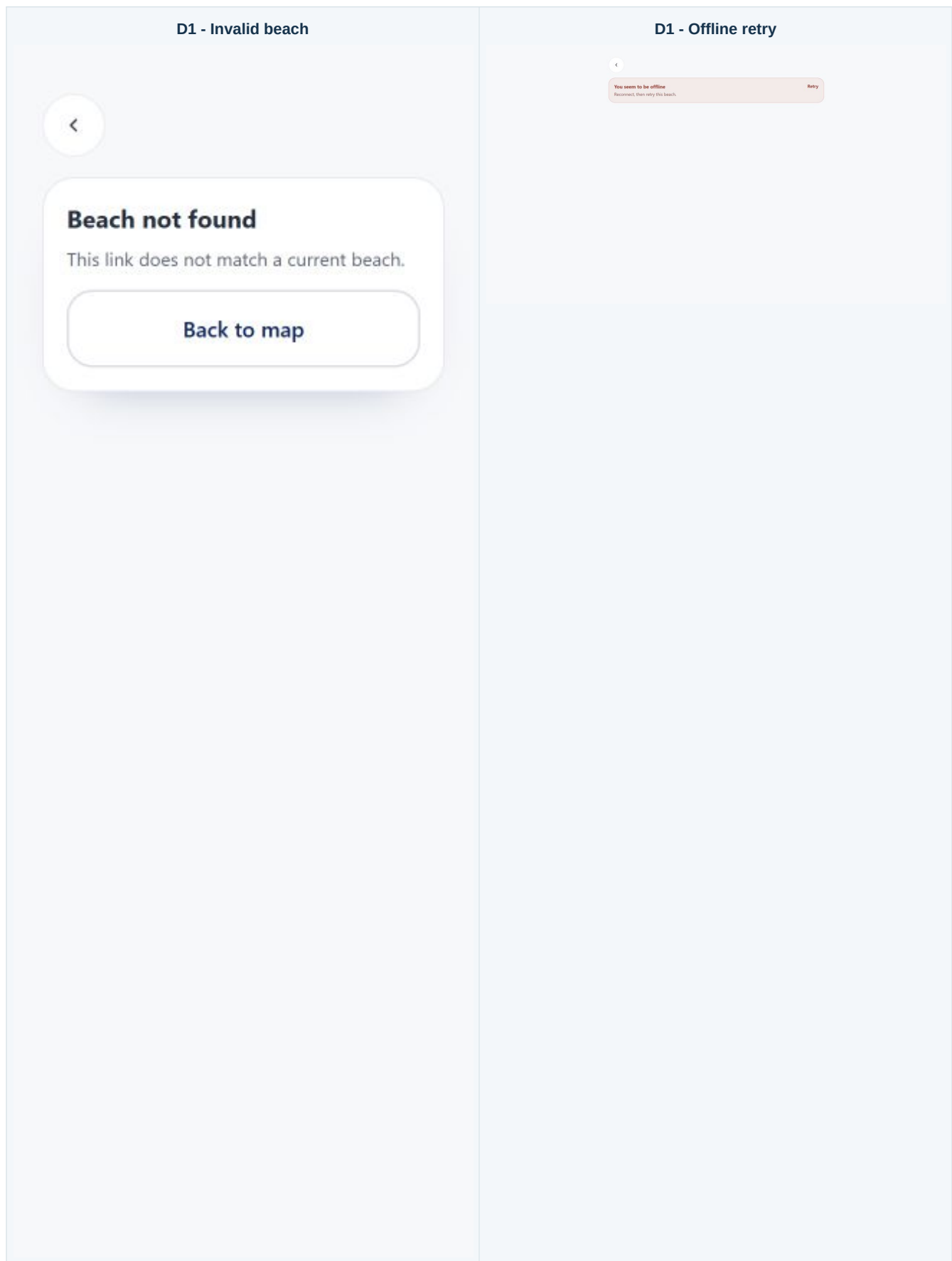
Synohum has no separate category in the app, so record it as Other.

CATEGORY & SIZE

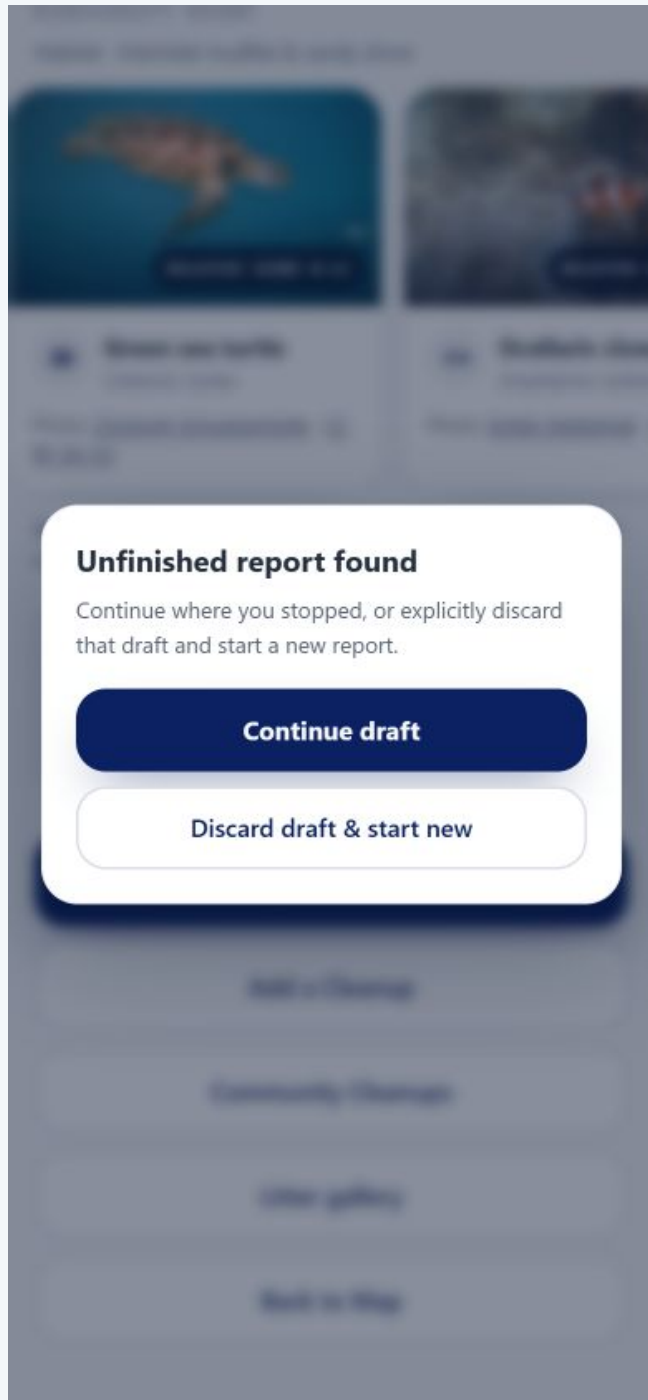
Category means litter type. Quantity uses Small, Medium, Large or Very Large. A new report with only Small bands is not counted as active evidence. Small is still used to record what remains after cleanup.

PHOTO PRIVACY

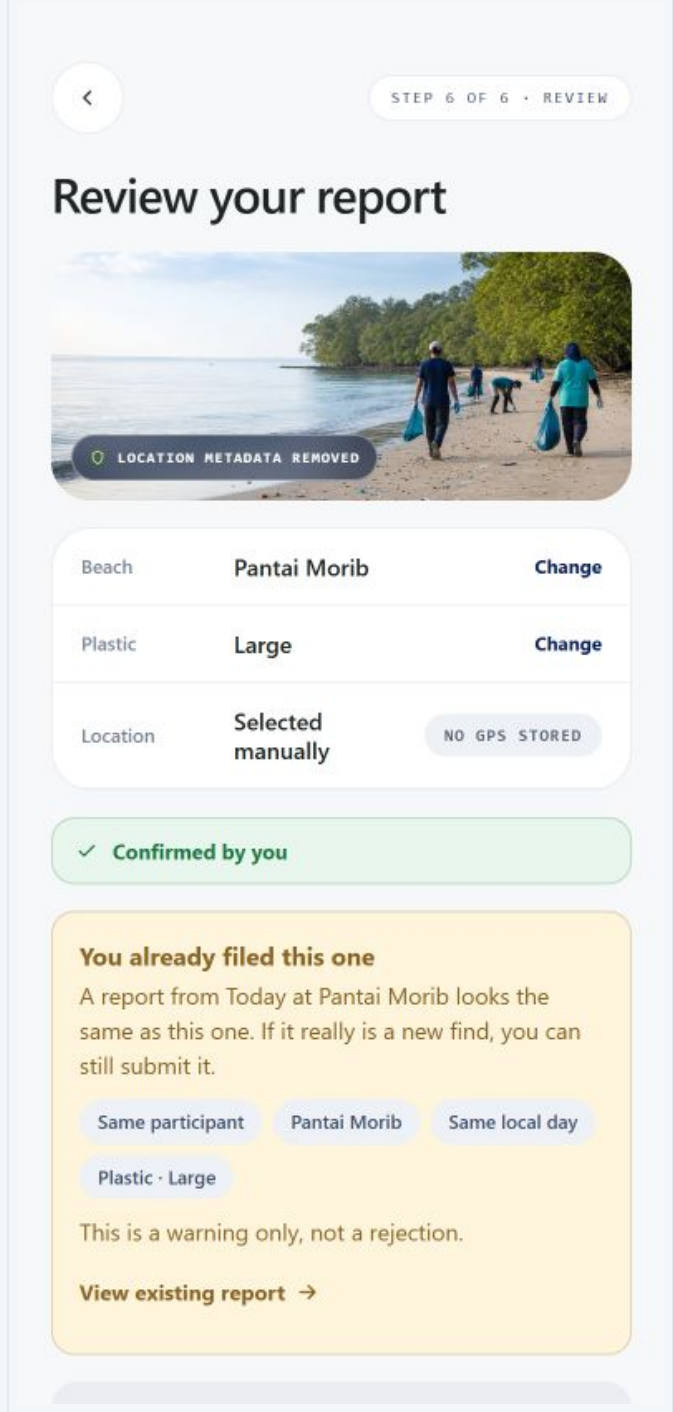
Ordinary report photos keep a protected photo



D3 - Draft choices, 390 x 844



D5 - Duplicate warning, 390 x 844

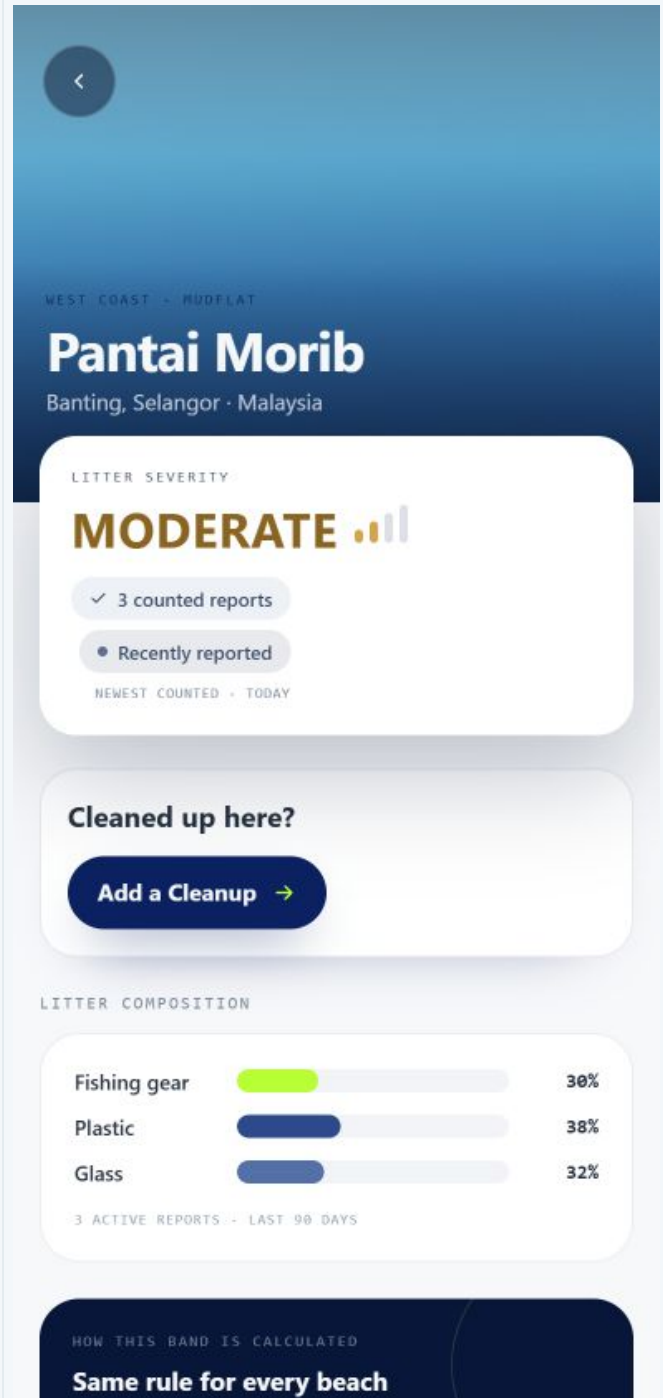


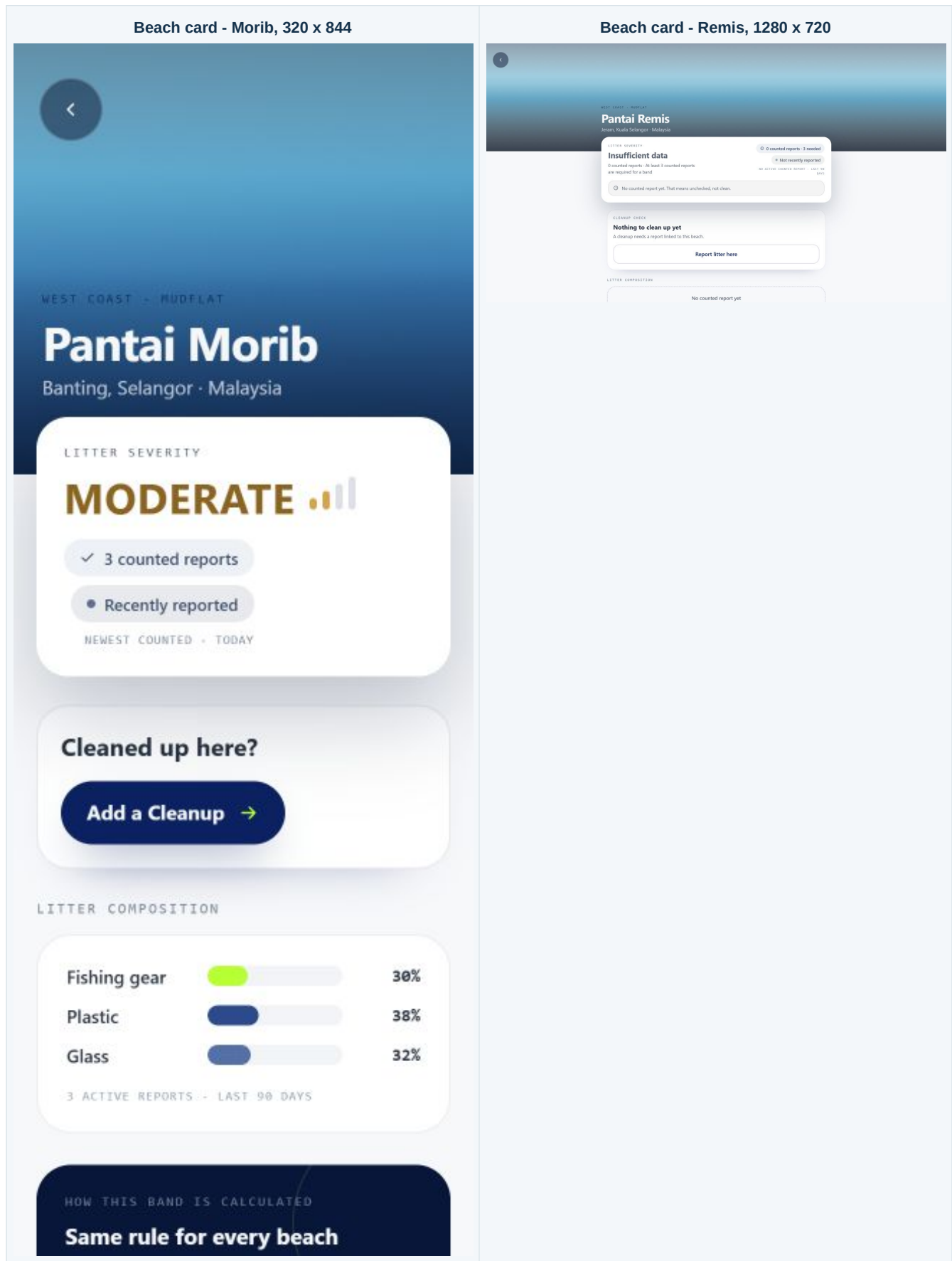
D7 - Four-digit participant ID	F1 - Boot shell under throttling
< Join in — no name needed Use your participant ID and recovery token. Get an ID Restore profile PARTICIPANT ID 1234 RECOVERY TOKEN Enter your recovery token Restore profile + Create a new profile 🛡️ Reports use your participant ID, not your name. Exact location stays private. Keep browsing without an ID	R5 RADAR SAMPAH Waking up the server...

F1 - Ready state after throttling



Beach card - Morib, 390 x 844





Beach card - Remis, 390 x 844



WEST COAST - MUDEFLAT

Pantai Remis

Jeram, Kuala Selangor · Malaysia

LITTER SEVERITY

Insufficient data

0 counted reports · At least 3 counted reports are required for a band

① 0 counted reports · 3 needed

• Not recently reported

NO ACTIVE COUNTED REPORT · LAST 90 DAYS

🕒 No counted report yet. That means unchecked, not clean.

CLEANUP CHECK

Nothing to clean up yet

A cleanup needs a report linked to this beach.

Report litter here

LITTER COMPOSITION

No counted report yet

Beach card - Remis, 320 x 844



WEST COAST - MUDEFLAT

Pantai Remis

Jeram, Kuala Selangor · Malaysia

LITTER SEVERITY

Insufficient data

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① 0 counted reports · 3 needed

• Not recently reported

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CLEANUP CHECK

Nothing to clean up yet

A cleanup needs a report linked to this beach.

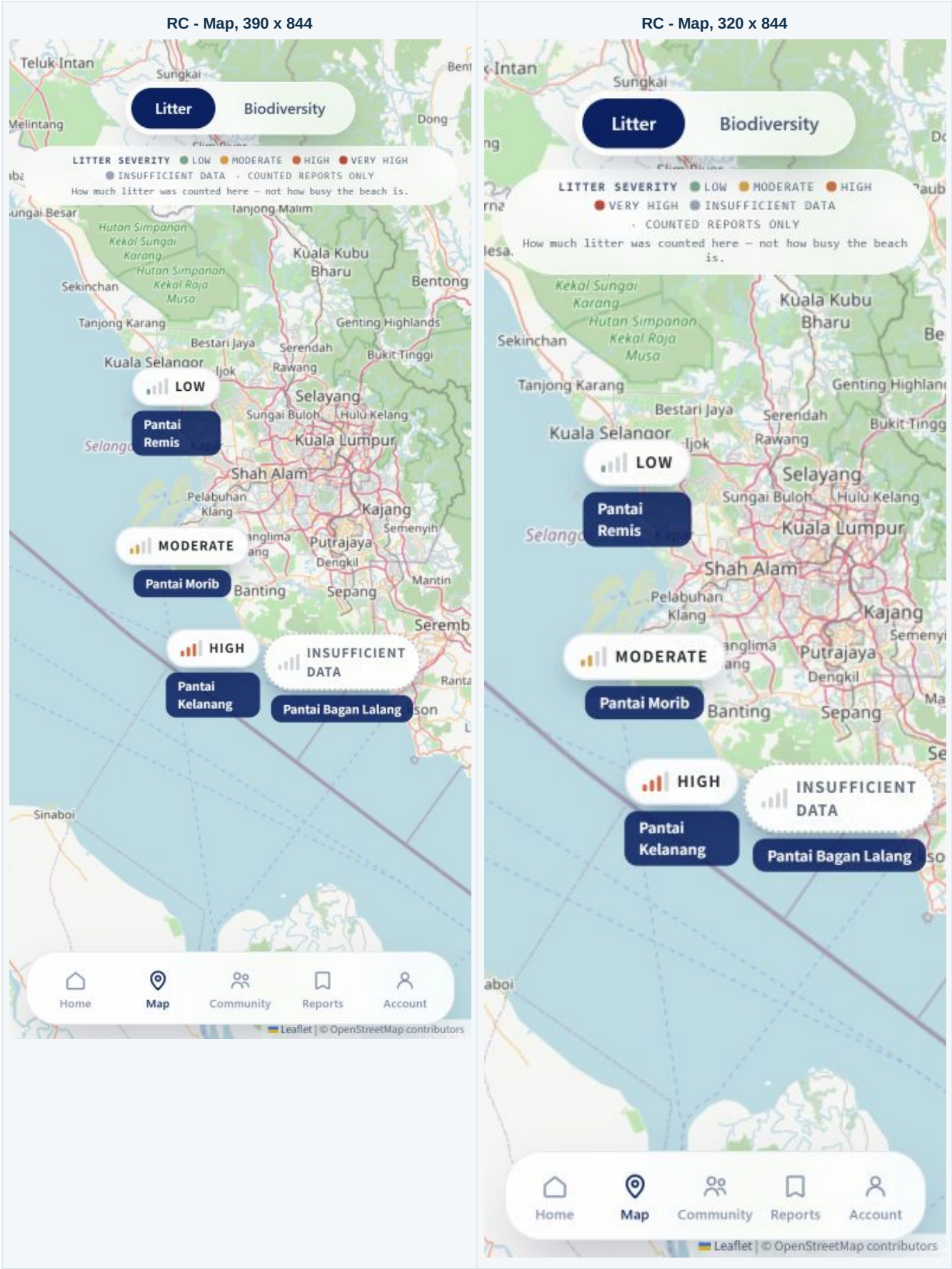
Report litter here

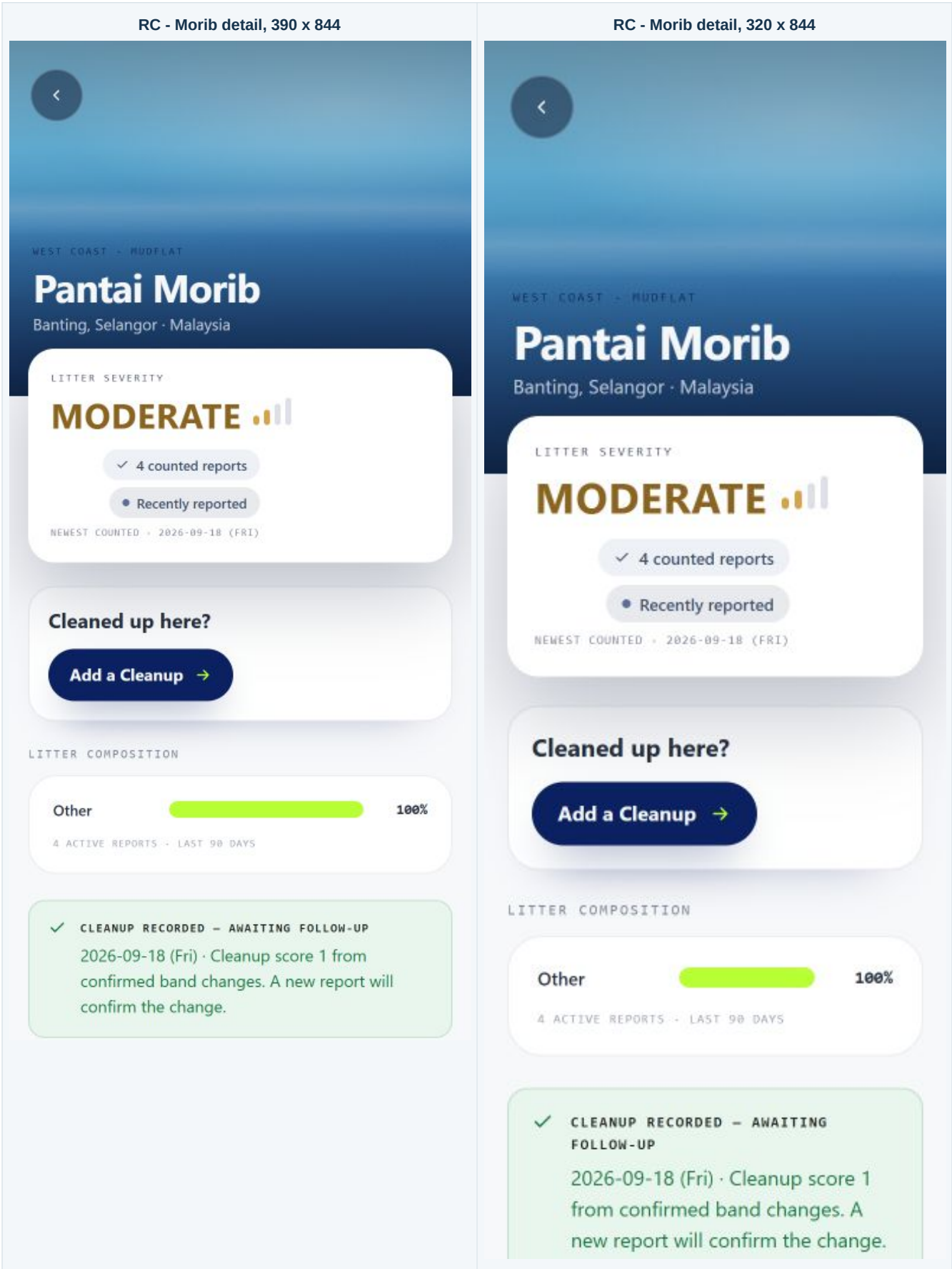
LITTER COMPOSITION

No counted report yet

Public release-candidate captures

Captured from the Render deployment matching application SHA 368212375a25b3443ac002e7526a657dca2e35f2.





RC - Method page, 390 x 844

<

SAME FOR ALL BEACHES

How the severity band is decided

Fixed litter rules. AI only suggests.

THE RULE

ONE CATEGORY

category weight × quantity level

ONE REPORT

highest category score

ONE BEACH

median of active counted report scores · latest 90 days

CATEGORY

QUANTITY

Fishing gear	1.00	Small	1
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RC - AI method page, 390 x 844

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- AI suggests a category and size.
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TRAINED WITH OPEN MARINE-LITTER PHOTOS FROM SEA AND TACO

WHAT IT CAN SUGGEST

Plastic

Fishing gear

Glass

Metal

Paper

Other

Styrofoam has no separate category in the app, so record it as Other.

CATEGORY & SIZE

Category means litter type. Quantity uses Small, Medium, Large or Very Large. A new report with only Small bands is not counted as active evidence; Small is still used to record what remains after cleanup.

PHOTO PRIVACY

Ordinary report photos keep a protected photo

